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Systems and methods for improving tolerance of delay and disruption of a control-to-data-plane interface in a software-defined network

Abstract

This disclosure provides systems and methods for improving tolerance of delay and disruption of a control-to-data-plane interface (CDPI) in a software-defined network. A system can include a plurality of moving nodes and an SDN controller communicatively coupled to the plurality of moving nodes. The SDN controller can be configured to send a first control message to a first moving node of the plurality of moving nodes according to a CDPI protocol. The first control message can include instructions for the first moving node to execute a modification of a physical network topology parameter. The SDN controller also can be configured to send a second control message to the first moving node according to the CDPI protocol. The second control message can include instructions for the first moving node to modify routing information stored by the first node based on the modification of the physical network topology parameter.

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Background/Summary

.Iadd.CROSS-REFERENCE TO RELATED APPLICATIONS .Iaddend.

(1) .Iadd.The present application is a reissue of U.S. Pat. No. 9,924,441, which is herein incorporated by reference as if fully set forth in this description. More than one reissue application has been filed for the reissue of U.S. Pat. No. 9,924,441. The reissue applications are application number are application Ser. No. 16/825,402 (the present application) filed Mar. 20, 2020; and application Ser. No. 16/825,467, filed Mar. 20, 2020, both of which are reissues of U.S. application

BACKGROUND

(2) Information can be transmitted over directional point-to-point computer networks, such as aerospace and other mobile networks. In such networks, links can be formed between pairs of nodes by aiming transceivers of each node pair towards each other. In some implementations, the network nodes may move over time, which can result in frequent disruption of the links in the network and can require frequent reconfiguration of the links. Typically, a control-to-data-plane interface (CDPI) in a software-defined network (SDN) requires reliable, low-latency links for transmitting control messages to network nodes. However, aerospace networks often do not have such reliable, low-latency links. Furthermore, CDPIs typically do not support messages to alter the physical topology of a directional point-to-point network.

SUMMARY

(3) In some aspects, the disclosure relates to a system for configuring a software-defined network (SDN). The system can include a plurality of moving nodes and an SDN controller communicatively coupled to the plurality of moving nodes. The SDN controller can be configured to send a first control message to a first moving node of the plurality of moving nodes according to a control-to-data-plane interface (CDPI) protocol. The first control message can include instructions for the first moving node to execute a modification of a physical network topology parameter, including at least instructions to cause the first moving node to aim a wireless transceiver towards a second moving node. The SDN also can be configured to send a second control message to the first moving node according to the CDPI protocol. The second control message can include instructions for the first moving node to modify routing information stored by the first moving node based on the modification of the physical network topology parameter.

(4) In some implementations, the SDN controller can be further configured to send the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter. In some implementations, the SDN controller can be further configured to send the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information. In some implementations, the SDN controller can be further configured to receive, from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter and the modification of the routing information stored by the first moving node have both been executed.

(5) In some implementations, the SDN controller can be further configured to receive, from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter has been executed, and to send the second control message responsive to receiving the confirmation message. In some implementations, the first control message can include instructions to cause the first moving node to modify at least one parameter associated with a transmitter of the first moving node. In some implementations, the at least one parameter associated with the transmitter of the first moving node can include at least one of a transmission power, a transmission frequency, and a modulation scheme.

(6) In another aspect, the disclosure relates to a method for configuring a software-defined network (SDN). The method can include sending, by an SDN controller communicatively coupled to a plurality of moving nodes, a first control message to a first moving node of the plurality of moving nodes according to a control-to-data-plane interface (CDPI) protocol. The first control message can include instructions for the first moving node to execute a modification of a physical network topology parameter, including at least instructions to cause the first moving node to aim a wireless transceiver towards a second moving node. The method also can include sending, by the SDN controller, a second control message to the first moving node according to the CDPI protocol. The second control message can include instructions for the first moving node to modify routing information stored by the first moving node based on the modification of the physical network

topology parameter.

(7) In some implementations, the method can include sending, by the SDN controller, the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter. In some implementations, the method can include sending, by the SDN controller, the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information. In some implementations, the method can include receiving, by the SDN controller from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter and the modification of the routing information stored by the first moving node have both been executed.

(8) In some implementations, the method can include receiving, by the SDN controller from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter has been executed, and sending, by the SDN controller, the second control message responsive to receiving the confirmation message. In some implementations, sending the first control message can include sending instructions to cause the first moving node to modify at least one parameter associated with a transmitter of the first moving node. In some implementations, sending the first control message can include sending instructions to cause the first moving node to modify at least one of a transmission power, a transmission frequency, and a modulation scheme.

(9) In another aspect, the disclosure relates to a non-transitory computer-readable medium having instructions encoded thereon which, when executed by one or more processors, cause the one or more processors to perform a method for configuring a software-defined network (SDN). The method can include sending, by an SDN controller communicatively coupled to a plurality of moving nodes, a first control message to a first moving node of the plurality of moving nodes according to a control-to-data-plane interface (CDPI) protocol. The first control message can include instructions for the first moving node to execute a modification of a physical network topology parameter, including at least instructions to cause the first moving node to aim a wireless transceiver towards a second moving node. The method also can include sending, by the SDN controller, a second control message to the first moving node according to the CDPI protocol. The second control message can include instructions for the first moving node to modify routing information stored by the first moving node based on the modification of the physical network topology parameter.

(10) In some implementations, the method can include sending, by the SDN controller, the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter. In some implementations, the method can include sending, by the SDN controller, the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information. In some implementations, the method can include receiving, by the SDN controller from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter and the modification of the routing information stored by the first moving node have both been executed.

(11) In some implementations, the method can include receiving, by the SDN controller from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter has been executed, and sending, by the SDN controller, the second control message responsive to receiving the confirmation message. In some implementations, sending the first control message can include sending instructions to cause the first moving node to modify at least one parameter associated with a transmitter of the first moving node. In some implementations, sending the first control message can include sending instructions to cause the first moving node to modify at least one of a transmission power, a transmission frequency, and a modulation scheme.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and related objects, features, and advantages of the present disclosure will be more fully understood by reference to the following detailed description, when taken in conjunction with the accompanying figures, wherein:
- (2) FIG. 1 is a block diagram of an example directional point-to-point computer network;
- (3) FIG. 2 is a block diagram of an example system for configuring a computer network;
- (4) FIG. 3A is a block diagram of an example computer network having a first topology;
- (5) FIG. 3B is a block diagram of the example computer network shown in FIG. 3A having a reconfigured network topology;
- (6) FIG. 3C is a flowchart of an example method for reconfiguring the computer network having the topology shown in FIG. 3A to achieve the topology shown in FIG. 3B;
- (7) FIG. 4A is a block diagram of an example computer network having a first topology;
- (8) FIG. 4B is a block diagram of the example computer network shown in FIG. 4A having a reconfigured network topology; and
- (9) FIG. 4C is a flowchart of an example method for reconfiguring the computer network having the topology shown in FIG. 4A to achieve the topology shown in FIG. 4B.
- (10) For purposes of clarity, not every component may be labeled in every figure. The drawings are not intended to be drawn to scale. Like reference numbers and designations in the various figures indicate like elements.

DETAILED DESCRIPTION

(11) The systems and methods of the disclosure relate to improving tolerance of delay and disruption of a control-to-data-plane interface (CDPI) in a software-defined network. Particularly, the disclosure describes an SDN controller and a CDPI that allows the SDN controller to send control messages to network nodes that may include instructions for the network node to aim a steerable beam antenna at another network node to form a communication link. The SDN controller also can send instructions to a network node relating to a power level to be used for transmitting information, for example based on a distance between the nodes forming the link, or other radio functions such as communication channels to be used for transmitting and receiving data. In addition, the SDN controller can send updated routing information to network nodes, as is done in traditional software-defined networks. In some implementations, the control messages sent by the SDN controller can include scheduling information, which can help to increase the tolerance of a software-defined aerospace network to delay and disruption of network links that may occur during a topology reconfiguration.

(12) FIG. 1 is a block diagram of an example directional point-to-point computer network **100**. The computer network **100** is a directional point-to-point computer network consisting of nodes mounted on various land- and air-based devices, some of which may change position with respect to other nodes in the network **100** over time. For example, the network **100** includes nodes associated with each of two land-based datacenters **105a** and **105b** (generally referred to as datacenters **105**), nodes associated with each of two ground stations **107a** and **107b** (generally referred to as ground stations **107**), and nodes associated with each of three airborne high altitude platforms (HAPs) **110a-110c** (generally referred to as HAPs **110**). As shown, the HAP **110a** is a blimp, the HAP **110b** is an airplane, and the HAP **110c** is a balloon. Arrows shown between a pair of nodes represent active communication links between the nodes. It should be understood that the network **100** as shown in FIG. 1 is illustrative only, and in some implementations the network **100** may include additional or different nodes. For example, in some implementations, the network **100** may include additional HAPs, which may be balloons, blimps, airplanes, unmanned aerial vehicles (UAVs), satellites, or any other form of high altitude platform. In some implementations, the

network **100** may serve as an access network for client devices such as cellular phones, laptop computers, desktop computers, wearable devices, or tablet computers. The network **100** also may be connected to a larger network, such as the Internet, and may be configured to provide a client device with access to resources stored on or provided through the larger computer network. In some implementations, the HAPs **110** can include wireless transceivers associated with a cellular or other mobile network, such as eNodeB base stations or other wireless access points, such as WiMAX or UMTS access points. Together, the HAPs **110** can form a wireless access network. The HAPs **110** can connect to the datacenters **105**, for example, via backbone network links or transit networks operated by third parties. The datacenters **105** may include servers hosting applications that are accessed by remote users as well as systems that monitor or control the components of the network **100**. The HAPs **110** can provide wireless access for the users, and can route user requests to the datacenters **105** and return responses to the users via the backbone network links.

(13) In some implementations, the network **100** can be an SDN that is controlled by an SDN controller, which may be located, for example, in one of the datacenters **105**. The nodes of the network **100** can be configured to communicate with one another using steerable wireless transceivers. The transceivers may be mounted to actuators that can be controlled to point in a desired direction. To form a link between two nodes, such as the node associated with the HAP **110a** and the node associated with the HAP **110a**, the transceivers of the respective nodes can be controlled to point in the direction of one another so that data can be sent and received between the nodes. In some implementations, the power of the signals transmitted by each transceiver can also be controlled to facilitate formation of the links in the network **100**. For example, nodes that are separated by a relatively large distance can be configured to operate at a higher power to compensate for the reduction in signal-to-noise ratio that occurs over the distance separating the two nodes. Nodes that are spaced nearer to one another may be controlled to operate at a relatively lower power so as to save power.

(14) As the HAPs **110** move with respect to one another and with respect to the datacenters **105** and ground stations **107** over time, some of the links shown in the block diagram of FIG. **1** may become infeasible. For example, the link between the ground station **107a** and the HAP **110a** may not be feasible when the path of the HAP **110a** brings the HAP **110a** into a position in which it is out of range of the ground station **107a**, or in which the earth is positioned between it and the ground station **107a**. Thus, due to the continuous movement of the HAPs **110**, the topology of the network **100** may require regular or irregular reconfiguration to maintain connectivity to satisfy provisioned network flows.

(15) Generally, traditional CDPI implementations such as OpenFlow do not include support for control messages that include instructions for steering wireless transceivers or for controlling transmitter parameters, such as transmission power, frequency, and modulation scheme. Accordingly, an SDN controller that uses a traditional CDPI may not be able to control the nodes of the network **100** to reconfigure the topology of the network **100**. Furthermore, most CDPIs require network links that are highly reliable and exhibit low latency. Due to the continuous movement of the HAPs **110**, as well as the relatively large distances between nodes in the network **100**, the network **100** may not exhibit the reliability and low latency required by typical CDPIs. As a result, it can be difficult to effectively manage the network **100** over time using a traditional SDN controller and CDPI. To address this issue, an SDN controller and CDPI can be configured to support control messages including instructions for network nodes to modify topology parameters, for example by reconfiguring their steerable wireless transceivers or by adjusting the power output of their transceivers. In addition, control messages can include instructions for a network node to execute topology and routing changes at specified times in the future, which can help to compensate for the delays or disruptions that may occur when attempting to form new links in a directional point-to-point network such as the network **100**.

(16) FIG. **2** is a block diagram of an example system **200** for configuring a computer network. The

system **300** includes an SDN controller **205** in communication with a network **207**. The SDN controller **205** communicates with the network **207** via a CDPI **206**. In some implementations, the network **207** may be or may include a directional point-to-point network having one or more nodes that move relative to other nodes over time, similar to the network **100** shown in FIG. **1**. The SDN controller **205** may be any type of computing device capable of sending control messages to the network **207** to configure the topology of the network **207** and to pass routing information to the nodes of the network **207**. Although the SDN controller **205** is shown as a component separate from the network **207**, it should be understood that in some implementations, the network controller **205** may be part of the network **207**. The network controller **205** includes a control message transceiver **210**, a topology management module **215**, a flow management module **220**, and a database **225**.

(17) In some implementations, the control message transceiver **210** can send control messages to the nodes of the network **207**, and can receive responses to control messages from the nodes. Control messages sent to a network node can include instructions for the network node to reconfigure topology parameters. For example, the control message transceiver **210** can send instructions to a node that causes the node to aim a steerable beam antenna at another network node to form a communication link. In some implementations, the control message transceiver **210** can send instructions to a network node relating to the functionality of the transmitter used by the node to communicate with other nodes, such as a power level, a frequency, or a modulation scheme to be used by the node for transmitting information. In addition, the control message transceiver **210** can send updated routing information to the nodes of the network **207**, as is done in traditional software-defined networks.

(18) As discussed above, reconfiguring the network **207** can cause temporary delay and disruption of network links as the nodes steer their transceivers to form new links. To increase the tolerance of the network **207** to such delay and disruption, the control message transceiver **210** can be configured to send control messages that include scheduling information. When the control message transceiver **210** instructs a node to break a first link and to instead establish a second link, there may be a significant delay between the breaking of the first link and the establishment of the second link. During this delay, the control plane (and the data plane) can be disrupted and will remain disrupted until the new link is formed and routing information for the new link is stored by the nodes of the network **207**. In some cases, this disruption could cause the network node to become stranded. To avoid stranding the network node, the control message transceiver **210** can instruct the node to establish the second link at a first scheduled time, and to update its routing information at a later time. The network node can receive these instructions and execute them autonomously according to the scheduling information associated with the instructions, even while the control plane is disrupted. An example of this is described further below in connection with FIGS. **3A-3C**.

(19) In some implementations, the control message transceiver **210** can be configured to wait for confirmation from a network node that a first network topology modification has been executed before sending instructions for a second network topology modification that is dependent on the first network control modification. For example, the control message transceiver **210** may instruct a node to form a new link by aiming its transceiver at another node. Then, before sending updated routing information corresponding to the new link, the control message transceiver **210** can wait for an acknowledgment from the node to confirm that the new link has been formed. After confirmation has been received, the control message transceiver **210** can instruct the network nodes to update their routing information accordingly. In some implementations, if the SDN controller does not receive confirmation that the new link was formed, a new topology can be selected instead. An example of this is described further below in connection with FIGS. **4A-4C**.

(20) The topology management module **215** can be configured to determine the current topology of the network **207**. In some implementations, the determination can be made based on information

received from the network nodes **207**. For example, the topology management module **215** can be configured to receive information from each node in the network **207** corresponding to the links currently formed by each node, and can aggregate this information to determine a complete topology of the network **207**, including any failed links that may exist within the network **207**. In some implementations, the this information can be stored in the database **225**.

(21) The flow management module **220** can be configured to determine all of the flows that are currently provisioned in the network **207**. In some implementations, the flow management module **220** can determine the provisioned flows based on information received from network applications or from a separate traffic engineering application. In some implementations, the flow management module **220** can aggregate the application data to determine the total amount of bandwidth required between each node pair in the network **207**. This information can be stored, for example, in the database **225**. Together, the topology management module **215** and the flow management module **220** can be configured to determine alternative topologies that may better facilitate processing of the provisioned network flows. When it is determined that a new topology may be desirable, the control message transceiver **210** can be configured to send control messages including instructions for the nodes of the network **207** to modify topology parameters and routing information to achieve the new topology. As discussed above, such control messages also may be associated with scheduling information that can help the network **207** to accommodate delay and disruption that may occur as a result of implementing the topology and routing changes.

(22) FIG. 3A is a block diagram of an example computer network having a first topology **301**. FIG. 3B is a block diagram of the example computer network shown in FIG. 3A having a reconfigured network topology **303**. FIG. 3C is a flowchart of an example method for reconfiguring the computer network shown in FIG. 3A to achieve the topology **303** shown in FIG. 3B. FIGS. 3A-3C are described together below.

(23) Referring now to FIG. 3A, a computer network topology **301** includes four nodes **305a-305d** (generally referred to as nodes **305**). The nodes **305** are coupled to one another by three links **308a-308c** (generally referred to as links **308**). In particular, the node **305a** is coupled to the node **305c** via link **308a**, the node **305a** is coupled to the node **305b** via the link **308b**, and the node **305b** is coupled to the node **305d** via the link **308c**. In this example, it is assumed that the node **305d** has only a single wireless transceiver and therefore can only form a single link with at most one other node **305**.

(24) Referring to FIG. 3B, the topology **303** includes the same nodes **305a-305d** that are shown in FIG. 3A, but the link **308c** that joins the node **305b** and the node **305d** in FIG. 3A is removed and replaced with the link **308d** which joins the nodes **305c** and **3085d** as shown in FIG. 3B. Because the node **305d** has only a single wireless transceiver, there is no way to reconfigure the topology **301** of FIG. 3A to achieve the topology **303** of FIG. 3B in a “hitless” manner (i.e., without temporarily disrupting the network while the transceiver of node **305d** is steered away from the node **305b** and towards the node **305c** to form the link **308d** with the node **305c**. As a result, the control plane will become disrupted immediately after the link **308c** is broken in order to establish the link **308d**, and it will remain disrupted until the link **308d** is formed and routing information corresponding to the new link **308d** has been updated on the network nodes **305**.

(25) In some implementations, the disruption described above could result in the node **305d** becoming stranded (i.e., unreachable from every other node **305**). For example, a first control message may be sent to the node **305d** to cause the node **305d** to steer its antenna away from the node **305b** and towards the node **305d** in order to form the new link **308d**. If a second control message is sent to the node **305d** before the link **308d** has been established (or to any other node along a path that may include the link **308d**), the second control message may not reach its intended destination and the routing modification associated with the second control message will never be executed, which can lead to stranding of the network node **305d**. To reduce the likelihood of such an event, control messages can include scheduling information specifying a time at which they

should be executed. The node **305d** can be configured to receive such control messages, and to execute them autonomously at the specified times even while the control plane is disrupted, which can help to reduce the likelihood of the node **305d** becoming stranded. This example is described further below in connection with FIG. **3C**.

(26) FIG. **3C** is a flowchart of an example method **309** for reconfiguring the computer network having the topology **301** shown in FIG. **3A** to achieve the topology **303** shown in FIG. **3B**. In brief overview, the method **309** includes sending, by an SDN controller, a first control message to at least one node of a plurality of moving nodes (stage **310**). The first control message includes instructions for the first node to execute a modification of a physical network topology parameter at a first time. The method **309** further includes sending, by the SDN controller, a second control message to the at least one moving node of the plurality of moving nodes (stage **320**). The second control message includes instructions for the first node to execute a modification of routing information at a second time, later than the first time. The method **309** further includes receiving, by the SDN controller, confirmation that the modification of the physical network topology parameter and the modification of the routing information have been executed (stage **330**).

(27) Referring again to FIGS. **3A-3C**, the method **309** includes sending, by an SDN controller, a first control message to at least one node of a plurality of moving nodes (stage **310**). The first control message includes instructions for the first node to execute a modification of a physical network topology parameter at a first time. In the above example, the first control message can be sent to the node **305d**. The first control message can include instructions for the node **305d** to steer its transceiver towards the node **305c**, and also can include instructions specifying a time at which the node **305d** should execute this topology modification.

(28) The method **309** includes sending, by the SDN controller, a second control message to the at least one moving node of the plurality of moving nodes (stage **320**). The second control message includes instructions for the first node to execute a modification of routing information at a second time, later than the first time. In the example above, the second control message also can be sent to the node **305d**. The routing modification instructions included in the second control message can correspond to the topology modification instructions included in the first control message. For example, the routing modification instructions included in the second control message can establish a new route from the node **305d** to the node **305c** via the link **308d**, and can delete the route from the node **305d** to the node **305b** via the link **308c**. In some implementations, the second control message can be sent before the topology **301** of FIG. **3A** has been modified (i.e., before the first time at which the topology modification instructions included in the first control message are to be executed), ensuring that the second control message can be received by the node **305d** before the control plane is disrupted. The second control message can include instructions specifying a second time at which the node **305d** should execute the routing modification, which can be later than the first time at which the node **305d** is to execute the topology modification. Scheduling the topology and routing modifications in this way can allow the node **305d** to complete the topology modification before attempting to update the corresponding routing information, and both modifications can be executed autonomously by the node **305d**, thereby reducing the likelihood of the node **305d** becoming stranded due to the disruption of the control plane that occurs when the link **308c** is broken. In some implementations, the difference between the first time and the second time can be based on the estimated time required for the node **305d** to execute the topology modification instructions included in the first control message.

(29) The method **309** includes receiving, by the SDN controller, confirmation that the modification of the physical network topology parameter and the modification of the routing information have been executed (stage **330**). In the above example, the node **305d** can send a confirmation message to the SDN controller indicating the modification instructions associated with the first and second control messages have been executed. The SDN controller can therefore determine that the network node **305d** is not stranded, and can then instruct the other nodes **305** in the network to update their

routing information accordingly (e.g., to account for the removal of the link **308c** and the establishment of the new link **308d**).

(30) FIG. **4A** is a block diagram of an example computer network having a first topology **401**. FIG. **4B** is a block diagram of the example computer network shown in FIG. **4A** having a reconfigured network topology **403**. FIG. **4C** is a flowchart of an example method for reconfiguring the computer network shown in FIG. **4A** to achieve the topology **403** shown in FIG. **4B**. FIGS. **4A-4C** are described together below.

(31) Referring now to FIG. **4A**, a computer network topology **401** includes five nodes **405a-405e** (generally referred to as nodes **405**). The nodes **405** are coupled to one another by three links **408a-408d** (generally referred to as links **408**). In particular, the node **405a** is coupled to the node **405b** via the link **408a**, the node **405b** is coupled to the node **405c** via the link **308b**, the node **405c** is coupled to the node **405d** via the link **408c**, and the node **405d** is coupled to the node **405e** via the link **405e**. In the example that follows, it is assumed that each of the nodes **405c**, **405d**, and **405e** have three wireless transceivers, and therefore each of these nodes **405c**, **405d**, and **405e** can form a link with up to three other nodes **405**.

(32) Referring to FIG. **4B**, the topology **403** includes the same nodes **405a-405e** and the same links **408a-408e** that are shown in FIG. **4A**, but also include the addition of the link **408e** that joins the node **405c** and the node **405e**. In some implementations, the new link **408e** can allow for more efficient transmission of information from node **405a** to node **405e**, for example by using the route **405a-405b-405c-405e** instead of the longer route **405a-405b-405c-405d-405e**.

(33) In this example, the new link **408e** can be established in a “hitless” manner, because each of the nodes **405c** and **405e** that are joined by the new link **408e** have at least one unused transceiver. As a result, unlike in the example discussed above in connection with FIG. **3A-3C**, it can be possible to establish the new link **408e** without disrupting the control plane. To ensure that this change is hitless, the SDN controller can confirm that the new link **408e** has been successfully established before sending control messages to the network nodes **405** to cause the nodes to update their routing information to include the new route **405a-405b-405c-405d-405e**, because use of the new route is dependent upon the establishment of the new link **408e**. This example is described further below in connection with FIG. **4C**.

(34) FIG. **4C** is a flowchart of an example method **409** for reconfiguring the computer network having the topology **401** shown in FIG. **4A** to achieve the topology **403** shown in FIG. **4B**. In brief overview, the method **409** includes sending, by an SDN controller, a first control message to at least one node of a plurality of moving nodes (stage **410**). The first control message includes instructions for the first node to execute a modification of a physical network topology parameter. The method **409** further includes receiving, by the SDN controller, confirmation that the modification of the physical network topology parameter has been executed (stage **420**), and sending, by the SDN controller responsive to receiving the confirmation, a second control message to the at least one moving node of the plurality of moving nodes (stage **430**). The second control message includes instructions for the first node to execute a modification of routing information.

(35) Referring again to FIGS. **4A-4C**, the method **409** includes sending, by an SDN controller, a first control message to at least one node of a plurality of moving nodes (stage **410**). The first control message includes instructions for the first node to execute a modification of a physical network topology parameter. In the above example, the first control message can be sent to the nodes **405c** and **405e**. The first control message can include instructions for each of the nodes **405c** and **405e** to steer their respective transceivers towards one another and to update any transceiver parameters, such as output power, frequency, and modulation scheme, to allow for the establishment of the new link **408e**.

(36) The method **409** includes receiving, by the SDN controller, confirmation that the modification of the physical network topology parameter has been executed (stage **420**). In the above example, each of the nodes **405c** and **405e** can be configured to send a response to the SDN controller

indicating that the nodes **405c** and **405e** have implemented the network topology changes that were specified in their respective control messages. After the SDN controller receives confirmation messages from both of the nodes **405c** and **405e**, the SDN controller can determine that the new link **408e** has been established and is ready to be used for routing information in the network.

(37) The method **409** includes sending, by the SDN controller responsive to receiving the confirmation, a second control message to the at least one moving node of the plurality of moving nodes (stage **430**). The second control message includes instructions for the first node to execute a modification of routing information. In the above example, the SDN controller can determine that the link **408e** is ready for use after it has received the confirmation messages from both of the nodes **405c** and **405e**. The SDN controller can then send control messages to the nodes **405c** and **405e** (as well as the other nodes **405**) indicating that the new, more efficient route **405a-405b-405c-405e** should be preferred over the old, less efficient route **405a-405b-405c-405d-405e**.

(38) In some implementations, if the SDN controller does not receive the confirmation messages from one of the nodes **405c** and **405e**, the SDN controller can determine that the new link **408e** has not been established. As a result, the SDN controller will not send updated routing information corresponding to any route that requires the link **408e**. In some implementations, the SDN controller can attempt to resend the control messages including instructions for each of the nodes **405c** and **405e** to steer their respective transceivers towards one another, and again wait for confirmation from both nodes that these changes have been executed before sending updated routing information to the nodes **405**.

(39) Implementations of the subject matter and the operations described in this specification can be implemented in digital electronic circuitry, or in computer software embodied on a tangible medium, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Implementations of the subject matter described in this specification can be implemented as one or more computer programs embodied on a tangible medium, i.e., one or more modules of computer program instructions, encoded on one or more computer storage media for execution by, or to control the operation of, a data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. The computer storage medium can also be, or be included in, one or more separate components or media (e.g., multiple CDs, disks, or other storage devices). The computer storage medium may be tangible and non-transitory.

(40) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled languages, interpreted languages, declarative languages, and procedural languages, and the computer program can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program may, but need not, correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, libraries, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(41) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., a field programmable gate array (“FPGA”) or an application specific integrated circuit (“ASIC”). Such a special purpose circuit may be referred to as a computer processor even if it is not a general-purpose processor.

(42) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular implementations of particular inventions. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(43) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(44) References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. The labels “first,” “second,” “third,” and so forth are not necessarily meant to indicate an ordering and are generally used merely to distinguish between like or similar items or elements.

(45) Thus, particular implementations of the subject matter have been described. Other implementations are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking or parallel processing may be used.

Claims

1. A system for configuring a software-defined network (SDN), the system comprising: a plurality of moving nodes; and .[an SDN controller.]. .Iadd.one or more processors .Iaddend.communicatively coupled to .[the.]. plurality of moving nodes, wherein the .[SDN controller is.]. .Iadd.one or more processors are .Iaddend.configured to: send a first control message to a first moving node of the plurality of moving nodes.[according to a control-to-data-plane interface (CDPI) protocol.], the first control message including instructions for the first moving node to execute a modification of a physical network topology parameter, including at least instructions to cause the first moving node to aim a wireless transceiver towards a second moving node; and send a second control message to the first moving node.[according to the CDPI protocol.], the second control message including instructions for the first moving node to modify routing information stored by the first moving node based on the modification of the physical network topology parameter.[...]. .Iadd., wherein the one or more processors are further configured to send the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter; and wherein the one or more processors are further configured to send the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information..Iaddend.
- .[2. The system of claim 1, wherein the SDN controller is further configured to send the first control message specifying a first future time at which the first moving node is to execute the

modification of the physical network topology parameter..].

.[3. The system of claim 2, wherein the SDN controller is further configured to send the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information..].

4. The system of claim 1, wherein the .[SDN controller is.] .Iadd.one or more processors are .Iaddend.further configured to receive, from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter and the modification of the routing information stored by the first moving node have both been executed.

5. The system of claim 1, wherein the .[SDN controller is.] .Iadd.one or more processors are .Iaddend.further configured to: receive, from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter has been executed; and send the second control message responsive to receiving the confirmation message.

6. The system of claim 1, wherein the first control message includes instructions to cause the first moving node to modify at least one parameter associated with a transmitter of the first moving node.

7. The system of claim 6, wherein the at least one parameter associated with the transmitter of the first moving node includes at least one of a transmission power, a transmission frequency, and a modulation scheme.

8. A method for configuring a software-defined network (SDN), the method comprising: sending, by .[an SDN controller.] .Iadd.one or more processors .Iaddend.communicatively coupled to a plurality of moving nodes, a first control message to a first moving node of the plurality of moving nodes .[according to a control-to-data-plane interface (CDPI) protocol.] , the first control message including instructions for the first moving node to execute a modification of a physical network topology parameter, including at least instructions to cause the first moving node to aim a wireless transceiver towards a second moving node; and sending, by the .[SDN controller.] .Iadd.one or more processors.Iaddend., a second control message to the first moving node.[according to the CDPI protocol.] , the second control message including instructions for the first moving node to modify routing information stored by the first moving node based on the modification of the physical network topology parameter.[...] .Iadd., wherein the one or more processors are further configured to send the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter; and wherein the one or more processors are further configured to send the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information..Iaddend.

.[9. The method of claim 8, wherein the method further comprises sending, by the SDN controller, the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter..].

.[10. The method of claim 9, wherein the method further comprises sending, by the SDN controller, the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information..].

11. The method of claim 8, wherein the method further comprises receiving, by the .[SDN controller.] .Iadd.one or more processors .Iaddend.from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter and the modification of the routing information stored by the first moving node have both been executed.

12. The method of claim 8, wherein the method further comprises: receiving, by the .[SDN controller.] .Iadd.one or more processors .Iaddend.from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter has been executed; and sending, by the .[SDN controller.] .Iadd.one or more processors.Iaddend., the second control message responsive to receiving the confirmation message.

13. The method of claim 8, wherein sending the first control message includes sending instructions

to cause the first moving node to modify at least one parameter associated with a transmitter of the first moving node.

14. The method of claim 13, wherein sending the first control message includes sending instructions to cause the first moving node to modify at least one of a transmission power, a transmission frequency, and a modulation scheme.

15. A non-transitory computer-readable medium having instructions encoded thereon which, when executed by one or more processors, cause the one or more processors to perform a method for configuring a software-defined network (SDN), the method comprising: sending, by .[an SDN controller.]. .Iadd.one or more processors .Iaddend.communicatively coupled to a plurality of moving nodes, a first control message to a first moving node of the plurality of moving nodes. [according to a control-to-data-plane interface (CDPI) protocol.], the first control message including instructions for the first moving node to execute a modification of a physical network topology parameter, including at least instructions to cause the first moving node to aim a wireless transceiver towards a second moving node; and sending, by the .[SDN controller.]. .Iadd.one or more processors.Iaddend., a second control message to the first moving node .[according to the CDPI protocol.], the second control message including instructions for the first moving node to modify routing information stored by the first moving node based on the modification of the physical network topology parameter.[...]. .Iadd., wherein the one or more processors are further configured to send the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter; and wherein the one or more processors are further configured to send the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information..Iaddend.

.[16. The non-transitory computer-readable medium 15, wherein the method further comprises sending, by the SDN controller, the first control message specifying a first future time at which the first moving node is to execute the modification of the physical network topology parameter..].

.[17. The non-transitory computer-readable medium of claim 16, wherein the method further comprises sending, by the SDN controller, the second control message specifying a second future time, later than the first future time, at which the first moving node is to modify the routing information..].

18. The non-transitory computer-readable medium of claim 15, wherein the method further comprises receiving, by the .[SDN controller.]. .Iadd.one or more processors .Iaddend.from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter and the modification of the routing information stored by the first moving node have both been executed.

19. The non-transitory computer-readable medium of claim 15, wherein the method further comprises: receiving, by the .[SDN controller.]. .Iadd.one or more processors .Iaddend.from the first moving node, a confirmation message indicating that the modification of the physical network topology parameter has been executed; and sending, by the .[SDN controller.]. .Iadd.one or more processors.Iaddend., the second control message responsive to receiving the confirmation message.

20. The non-transitory computer-readable medium of claim 15, wherein sending the first control message includes sending instructions to cause the first moving node to modify at least one parameter associated with a transmitter of the first moving node.

21. The non-transitory computer-readable medium of claim 20, wherein sending the first control message includes sending instructions to cause the first moving node to modify at least one of a transmission power, a transmission frequency, and a modulation scheme.
